LLM-assisted specimen; source attributions retain their original authorship. See PROVENANCE.md and the tagged Clean sources.

Context rendering options

Both sections below come from the same Clean source and have the same Lean interpretation.

1 Expanded contexts

Every result repeats its required declarations and all inherited assumptions.

Result 1 (determinant product). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(A)\,\det(B).$$

Result 2 (determinant transpose). [Mathlib]
Let n be a natural number. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Then

$$\det(A^{\mathsf{T}}) = \det(A).$$

Result 3 (determinant product swap). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(B\,A).$$

Result 4 (determinant left swap). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,(B\,C)) = \det(B\,(A\,C)).$$

Result 5 (determinant right swap). [Mathlib]
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B\,C) = \det(A\,C\,B).$$

2 Referenced contexts

The context is printed once, and each result explicitly cites it.

Context 1 (complex square matrices).
Let n be a natural number. Let A, B and C be $n \times n$ matrices over $\mathbb{C}$.

Result 6 (determinant product). [Mathlib]
Using Context (1). Then

$$\det(A\,B) = \det(A)\,\det(B).$$

Result 7 (determinant transpose). [Mathlib]
Using Context (1). Then

$$\det(A^{\mathsf{T}}) = \det(A).$$

Result 8 (determinant product swap). [Mathlib]
Using Context (1). Then

$$\det(A\,B) = \det(B\,A).$$

Result 9 (determinant left swap). [Mathlib]
Using Context (1). Then

$$\det(A\,(B\,C)) = \det(B\,(A\,C)).$$

Result 10 (determinant right swap). [Mathlib]
Using Context (1). Then

$$\det(A\,B\,C) = \det(A\,C\,B).$$